

UNIT 13

PHYSICAL OPTICS



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OPTICS:

"Study of electromagnetic waves is called optics."

GEOMETRICAL OPTICS:

"Geometrical optics has to do with the principles that govern the imageforming properties of lenses, mirrors, and other devices that make use of light."

PHYSICAL OPTICS:

"Physical optics deals primarily with the nature and properties of light itself."

ELECTROMAGNETIC WAVES:

"Waves which consist of varying electric and magnetic fields and do not required medium for propagation are called electromagnetic waves."

ELECTROMAGNETIC SPECTRUM OF LIGHT:**• RADIO WAVES:**

Range: Wave length range of radio wave is typically from 0.1 m to 3×10^4 m approximately.

Discovered: Heinrich Hertz proved the existence of radio waves in the late 1880s.

Production: Stars are natural emitters of radio waves. Radio waves can be artificially generated by oscillating the current in the transmitting antenna.

• MICROWAVES:

Range: Wave length range of microwave is typically from 0.3 m to 30 cm approximately.

Discovered: The microwave was invented in 1945 by Raytheon engineer Percy Spencer.

Production: These are usually generated inside the specialized oven by an electron tube.

• INFRARED WAVES:

Range: Wave length range of infrared wave is typically from 0.1 mm to 10^{-4} m approximately.

Discovered: Sir William Herschel discoverer of infrared radiation in 1800.

Production: IR radiations radiated or absorbed by molecules when they change their rotational-vibrational movements.

• VISIBLE WAVES OR LIGHT WAVES:

Range: Wave length range of visible wave is typically from 380 nm to 700 nm approximately.

Discovered: Visible Light Spectrum was discovered by Isaac Newton in the 17th century.

Production: Visible waves or Light is produced by excited electrons dropping back to a lower energy state and releasing photons.

• ULTRAVIOLET WAVES:

Range: Wave length range of UV wave is typically from 100 nm to 400 nm approximately.

Discovered: On 22 February 1801, Johann Wilhelm Ritter discovered UV radiation in Jena.

Production: Very hot object objects such as the sun, emit the radiation known as ultraviolet waves.

The Ultraviolet is also produced by passing an electric current through mercury vapors in the tube.

• X-RAYS:

Range: Wave length range of X-Ray is typically from 0.01 m to 10 nm approximately.

Discovered: Wilhelm Roentgen, Professor of Physics in Wurzburg, Bavaria, discovered X-rays in 1895

Production: X-Rays are produced when fast moving electrons lose their energy quickly. The radiations are produced when a beam of fast moving electrons hits the metal target.

• GAMMA RAYS:

Range: Wave length range of Gamma rays is typically less than 10^{-11} m approximately.

Discovered: Paul Villard, a French chemist and physicist, discovered gamma radiation in 1900.

Production: Gamma rays come from radioactive materials. They are produced when the nuclei of unstable atoms decay into a stable nucleus or lose energy. They tend to have high energy than x-rays.

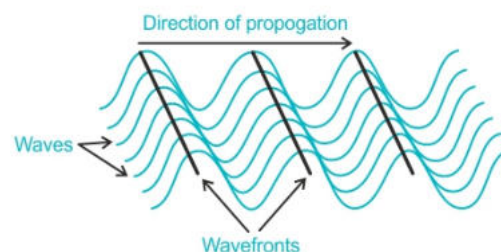
Wave front:

"A wave front is an imaginary surface representing corresponding points of a wave that vibrate in same phase."

Types of wave front:

There are three types of wave fronts.

- Plane wave front
- Spherical wave front
- Cylindrical wave front

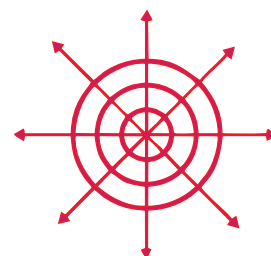


Spherical Wave front:

"When the point source is an isotropic medium, sending out waves three dimensions, the wavefronts are spheres centered on the source. Such a wavefront is called a spherical wavefront."

Examples of Spherical Wavefronts

- Electromagnetic waves in a vacuum form a spherical wavefront.
- The concentric circles' formation when the stone is dropped in water.



Cylindrical Wavefront:

"When the light source is linear, we obtain a cylindrical wavefront. In a cylindrical wavefront, all the points equidistant from the linear source lie on the surface of a cylinder."

Examples of Cylindrical Wavefronts

- When light from a lens falls on another lens and converges at a given point. As they bend and converge, it takes the form of the cylinder at that particular point.

Example:

- Surface of soap bubble.
- Oil layer on water.

Colors in thin film:

When light fall on this thin film various colours can be seen. These colours are the result of interference of light waves reflected from lower and upper surface of films.

Explanation:

Consider a thin film of refractive index 'n' and thickness 't'. Let a ray AB of monochromatic light of wavelength 'λ' be incident on the film.

This ray will be partially reflected along bc and partially refracted along bd and refract along dg.

The refracted ray 'de' will emerge out along 'ef'. The rays bc and ef interfere with each other producing on phase relationship.

It is clear from the figure that ray ef covers greater path as compared to ray bc.

The path difference between ray bc and ef depends upon the angle of incidence, refractive index of the film and thickness of the film. In a refracting medium wave length of light changes.

If 'λn' be the wave length of the light in the thin film then it is given by:

$$\lambda n = \frac{\lambda}{n}$$

For nearly normal incident the path difference between interfering rays bc and ef is "2t". Moreover, ray upon reflection from denser medium undergoes a phase change of 180° i.e. crest is reflected as trough and vice versa while no phase change occurs when it is reflected from a denser medium. Under these situations the conditions for constructive interference changes and can be given by:

$$2t = \left(m + \frac{1}{2}\right) \lambda n$$

Substituting the value of λn we get,

$$2t = \left(m + \frac{1}{2}\right) \frac{\lambda}{n}$$

$$2tn = \left(m + \frac{1}{2}\right) \lambda \dots\dots\dots (\text{maxima})$$

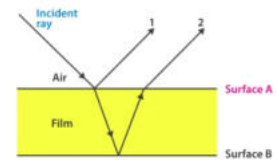
Where m = 0, 1, 2, 3,

For destructive interference we can write,

$$2tn = m\lambda \dots\dots\dots (\text{minima})$$

Where m = 0, 1, 2, 3,

'm' is called the order of image.



UNIT 14

COMMUNICATION



COMMUNICATION CHANNELS:

"The communication channel is the medium chosen by the sender (transmitter) for the transmission of signals to the receiver."

Categories of communication channels:

The communication channels can be divided into two categories: cable and broadcast. These categories are further divided into the following main components.

Cable:

- Twisted pair cable.
- Coaxial cable.
- Optical fiber cable.

Broadcast:

- Radio or infrared link.
- Microwave link.
- Satellites.

MERITS OF CHANNEL OF COMMUNICATION:**Twisted Wire Pair:**

- Easy to implement and low cost for short distances. Breakage in a segment doesn't affect the whole network. Less vulnerable to interference.

Coaxial Cable:

- Suitable for analog and digital data transmission, higher bandwidth, and cost effective compared to fiber optic cables.

Optical Fiber:

- High bandwidth and speed, cost-effective for long distances, and minimal signal loss compared to copper wires.

Microwave Radio Systems:

- Transmit large data volumes over long distances without physical cables, using repeaters. Lower construction costs than other transmission methods.

SATELLITE COMMUNICATION:

- It is versatile and location-independent, providing mobile and wireless communication. A single satellite can cover wide areas, entire countries, or regions. It is easy to install and manage ground stations. Used for audio, data, video, internet, and GPS applications, it has various uses like forecasting, broadcasting, military intelligence, navigation, global mobile communication, and connecting remote areas.

Modulation:

"The process of superimposing a low-frequency signal on a high-frequency carrier signal is called modulation."

Representation of carrier wave:

Mathematically a carrier wave is represented by a sinusoidal wave.

$$V(t) = A \sin(2\pi ft + \phi)$$

Representation of modulating wave:

$$V(t) = B \sin(2\pi ft)$$

Types of modulation:

Modulation is largely divided into two major types:

1. Amplitude modulation.
2. Angle modulation.

The angle modulation is further divided into two main types:

1. Frequency modulation.
2. Phase modulation.

AMPLITUDE MODULATION:

"It is the type of modulation in which the amplitude of the carrier signal is varied in proportion to the message signal whereas the frequency and phase of the carrier are constant."

Expression for amplitude modulated wave:

We have carrier wave and modulating signal,

$$m(t) = A_m \sin \omega_m t \text{ and } c(t) = A_c \sin \omega_c t \quad \dots\dots\dots 1$$

$m(t)$ = modulating signal and $c(t)$ = carrier wave.

A_m and A_c are Amplitude of modulating signal and carrier wave respectively in Amplitude modulation. Amplitude-modulated wave $C_m(t)$ will be

$$C_m(t) = (A_c + A_m \sin \omega_m t) \sin \omega_c t \quad \dots\dots\dots 2$$

This is the general form of amplitude modulated wave.

Where,

$$A = A_c + A_m \sin \omega_m t = \text{Amplitude of the modulated wave}$$

$$\sin \omega_c t = \text{Phase of modulated wave}$$

$$C_m(t) = A_c (1 + A_m/A_c \sin \omega_m t) \sin \omega_c t$$

$$= A_c \sin \omega_c t + A_m/A_c A_c \sin \omega_m t \sin \omega_c t$$

Where,

$$A_m/A_c = \mu = \text{modulation index}$$

$$C_m(t) = A_c \sin \omega_c t + A_c \mu \sin \omega_m t \sin \omega_c t$$

This is the general form of the amplitude modulated wave.

FREQUENCY MODULATION:

Frequency modulation is a technique or a process of encoding information on a particular signal (analogue or digital) by varying the carrier wave frequency in accordance with the frequency of the modulating signal.

Expression for frequency modulated wave:

As we know from amplitude modulation, we need two sine (or) cosine waves for modulation.

$$m(t) = A_m \cos (\omega_m t) \text{ and } c(t) = A_c \cos (\omega_c t) \quad \dots\dots\dots 1$$

or

$$m(t) = A_m \cos (2\pi f_m t) \text{ and } c(t) = A_c \cos (2\pi f_c t)$$

Then frequency modulated wave will be;

$$f_m(t) = f_c + k A_m \cos (2\pi f_m t) \quad \dots\dots\dots 2$$

$$f_m(t) = f_c + k m(t)$$

Where,

$f_m(t)$ = is frequency modulated wave

$f_c \rightarrow$ frequency of the carrier wave

$m(t) \rightarrow$ modulating signal and $k \rightarrow$ proportionality constant.

• Simplicity:

AM modulation is relatively simple to implement, making it cost-effective and widely used in broadcasting.

• Efficient use of bandwidth:

- **Compatibility:**

AM receivers can pick up weak signals, making it suitable for reception in areas with weaker signals or during atmospheric disturbances.

- **Immunity to sudden interference:**

AM signals are less affected by sudden noise or interference, allowing for clearer reception during temporary disturbances.

Advantages of FM (Frequency Modulation):

- **Better sound quality:**

FM provides better sound quality compared to AM, making it ideal for broadcasting music and high-fidelity audio.

- **Noise immunity:** FM is less susceptible to noise and static interference, resulting in clearer and more consistent reception.

- **Wider frequency range:** FM has a wider frequency range, allowing for more channels and better transmission of stereo signals.

- **Higher signal-to-noise ratio:** FM provides a higher signal-to-noise ratio, enhancing the overall signal quality and reducing the impact of background noise. In summary, AM is advantageous for simplicity, cost-effectiveness, and long-range communication.

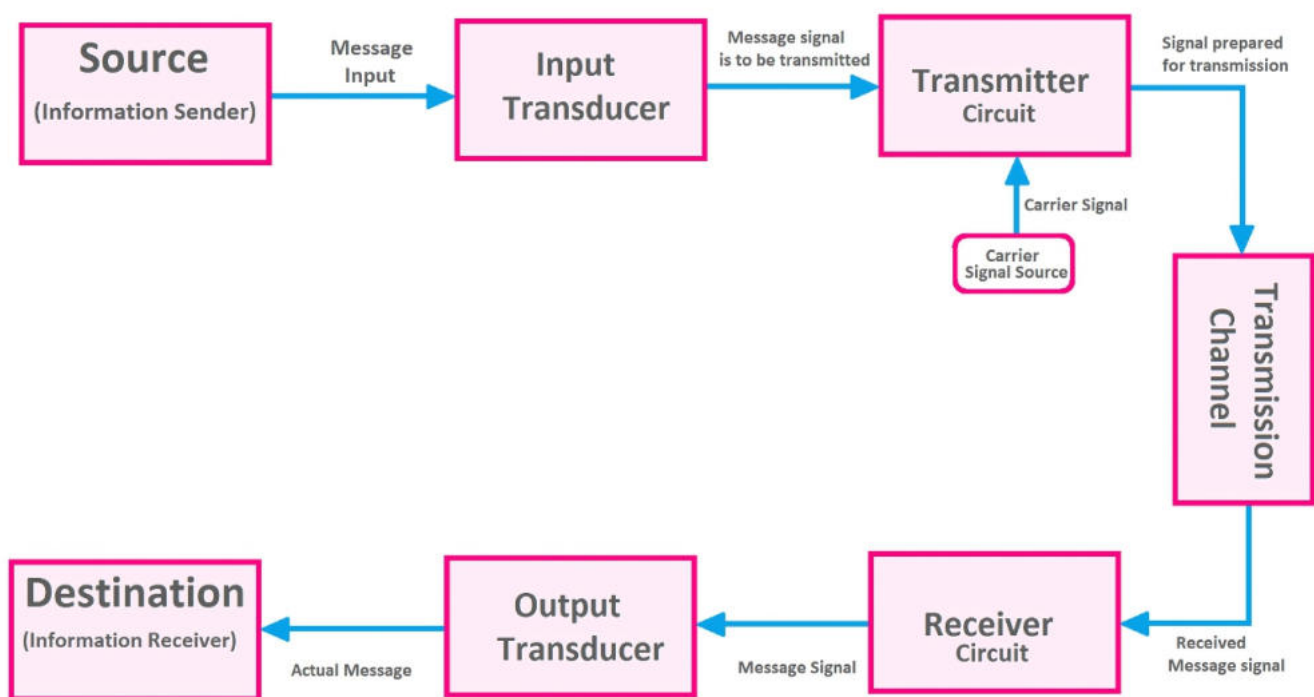
- **Superiority:** FM offers superior sound quality, noise immunity, and a wider frequency range, making it suitable for broadcasting high-quality audio and music.

- **Bandwidth:** Bandwidth specifically refers to the capacity at which a network can transmit data. For example, if the bandwidth of a network is 40 Mbps, it implies that the network cannot transmit data faster than 40 Mbps in any given case.

- **Digital Communication System:**

- It is the system in which messages generated by the source which are generally in analog form are converted to digital format and then transmitted.

- A digital communication system consists of six blocks:



Advantages of transmission of data in digital form:

- **Performance:** Analog signals suffer from distortion and noise even if they are small. In digital signals transmission can be made perfect due to less noise and distortion.
- **Compression and security:** Source and error control codes does not provide maximum accuracy, fidelity, compression, and security to a digital transmission that cannot be attained with analogue signals. Digital signals can be saved and recovered more easily than analog signals.
- **Multiplexing:**
 - Multiplexing of several different digital signals can be done easily. For example, combining digital signals using Time Division Multiplexing (TDM) is much easier than combining analog signals using Frequency Division Multiplexing (FDM).
- **Storage:**
 - In analog signal storage quality cannot be sustained over time. Whereas, Digital signals can be stored and retrieved more accurately and inexpensively by the use of transistors and error control codes.
- **Signal Processing:**
 - Signal Processing of Digital signals can be achieved in a simple and flexible way by using different program updates. Furthermore, we can change protocols and algorithms as per our requirement. This provides much more flexibility as compared to analog signal processing.
- **Reconstruction:**
 - Digital signals can be faultlessly carried over larger distances by using repeaters. However, analog signals have become gradually weaker along the channel due to severe noise and distortion. This behavior limits the range of transmission of analog signals governed by the input power.
- **Cost:**
 - Digital communication systems depend on computing devices like processors and very large-scale integrated (VLSI) circuits that increasingly benefit from Moore's law: the number of transistors in an integrated circuit (IC) has been increased rapidly and almost doubled about every two years. This leads to cost and performance advantages over a long period of time.
- **Secrecy:**
 - Cross-talk is rarely happening in digital communication due to encryption and compression of data in digital transmission to maintain the secrecy of the information.
- **Quality of Copies:**
 - Quality of copies in analog transmission is not good as compared to its original while due to error free digital transmission, copies can be made definitely.
- **A/D Conversion:**
 - "The method which is used to convert analog signals into digital is called analog to digital (A/D) conversion."
- **Analog to Digital Converter (ADC):**
 - "The device which is used to convert analog signals into digital signals is called ADC."
- **Steps for A/D Conversion:**
 - There are three processes involved in conversion of analogue to digital signals:

Sampling:

This is a process of inspecting the value (voltage) of an analog signal at regular time intervals. The time between samples is referred to as the **sample period** (T , in seconds), and the number of samples taken per second is referred to as the **sample frequency** (f_s , in samples/second or Hz). The receiver must convert the bits it receives into sample values, and then recreate what it thinks the analog signal looks like from the samples.

One must sample faster than the **NY-Quist sampling rate** (also called the NY-Quist rate), f_n , given by the formula $f_n = 2f_{\max}$, where $f_{\max}$ is the highest frequency component of the analog signal. To avoid distortion of your signal $f_s > f_n$.

Quantization:

Quantizing is the process of mapping the sampled analog voltage values to discrete voltage levels, which are then represented by binary numbers (bits). Converting a sample value from infinite possibilities to one of a finite set of values is called quantization. These values are referred to as quantization levels. An N -bit A/D converter has 2^N quantization levels and outputs binary words of length N . In general, the A/D converter will partition a range of voltage from $V_{\min}$ to some $V_{\max}$ into 2^N voltage intervals for each of size q volts, where:

$$q = \frac{V_{\max} - V_{\min}}{2^N}$$

Encoding:

After quantization the samples are converted to N -Bit binary code words. For the first sample point at time 0, the voltage is 0.613 V, which means that sample is assigned a binary value of 110. The A/D converter creates a voltage signal that represents these bits. Bit rate is the speed of transfer of data given in number of bits per second.

Digital to Analog Conversion (D/A Conversion):

The analog signal is reconstructed by converting the N -Bit digital words into appropriate quantization levels, and this voltage is "held" for one sample period, creating a stair-step type signal. Quantization error can be reduced by increasing the number of bits N for each sample. This will make the quantization interval smaller.